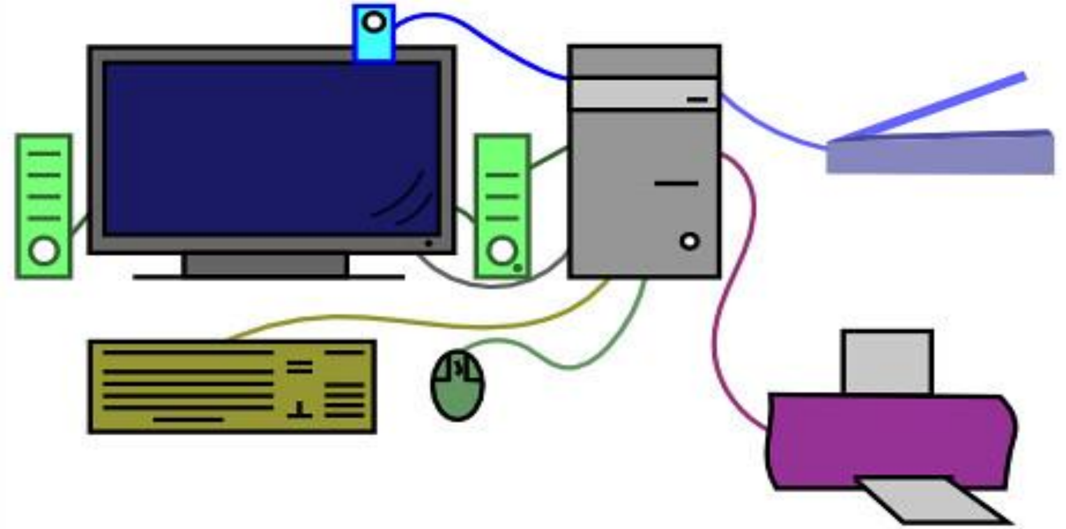

Basic Computer Organization

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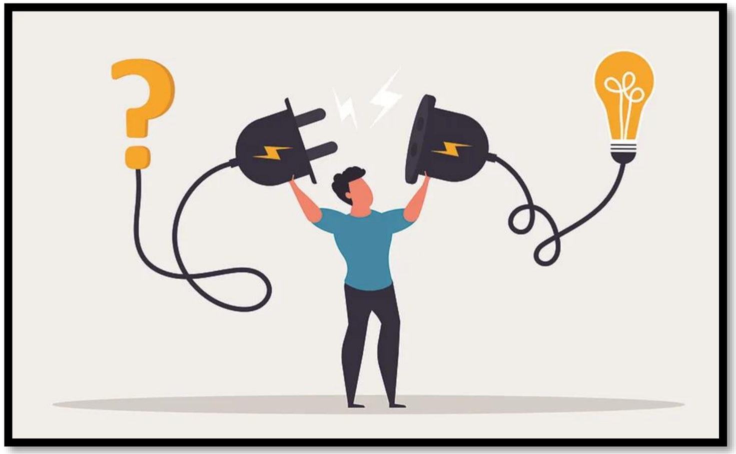
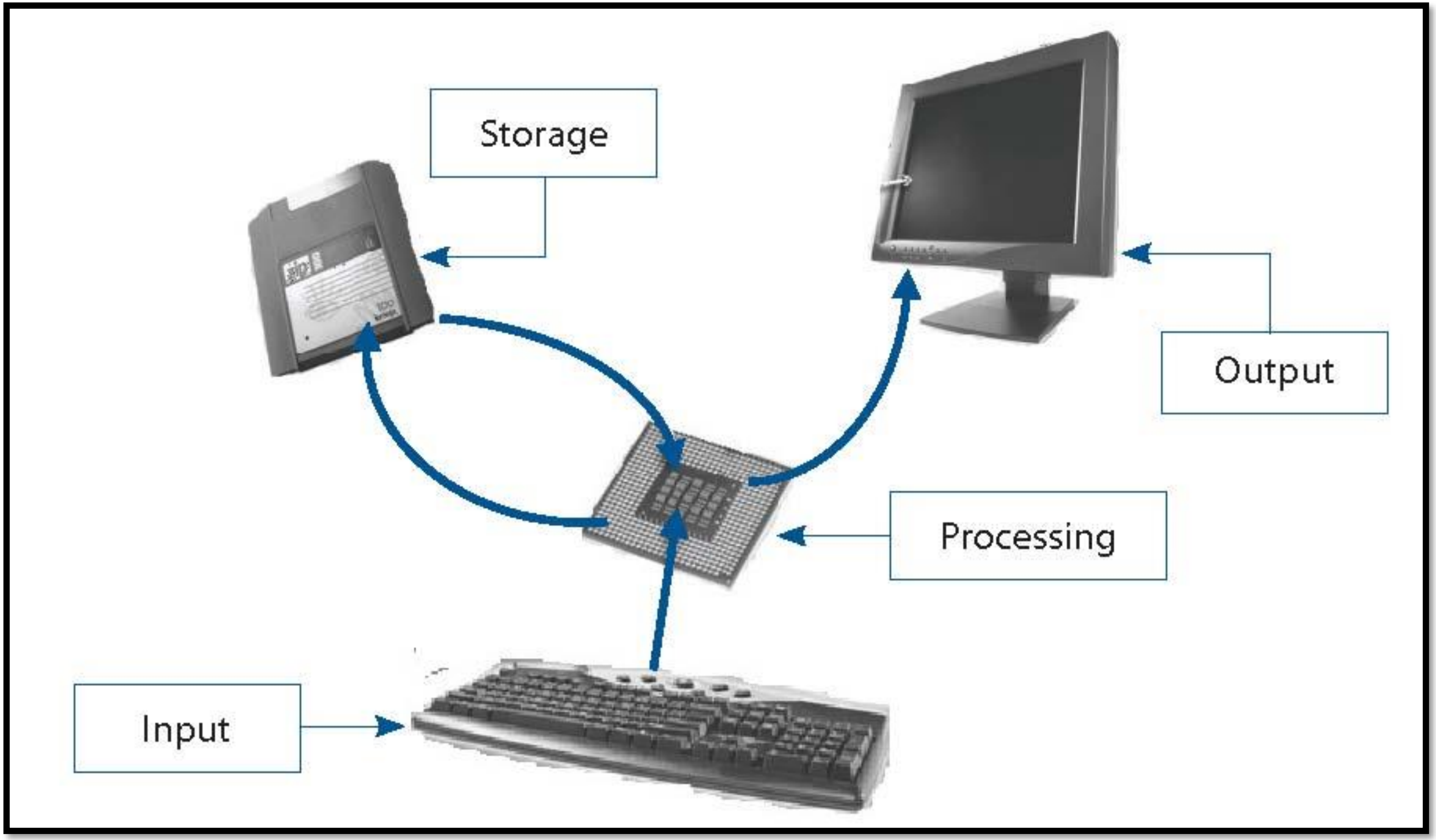
Learning Objectives

- **In this lecture you will learn about:**
  - ✓ Basic operations performed by all types of computer systems
  - ✓ Basic organization of a computer system
  - ✓ Input unit and its functions
  - ✓ Output unit and its functions
  - ✓ Storage unit and its functions
  - ✓ Types of storage used in a computer system
  - ✓ Arithmetic Logic Unit (ALU)
  - ✓ Control Unit (CU)
  - ✓ Central Processing Unit (CPU)
  - ✓ Computer as a system



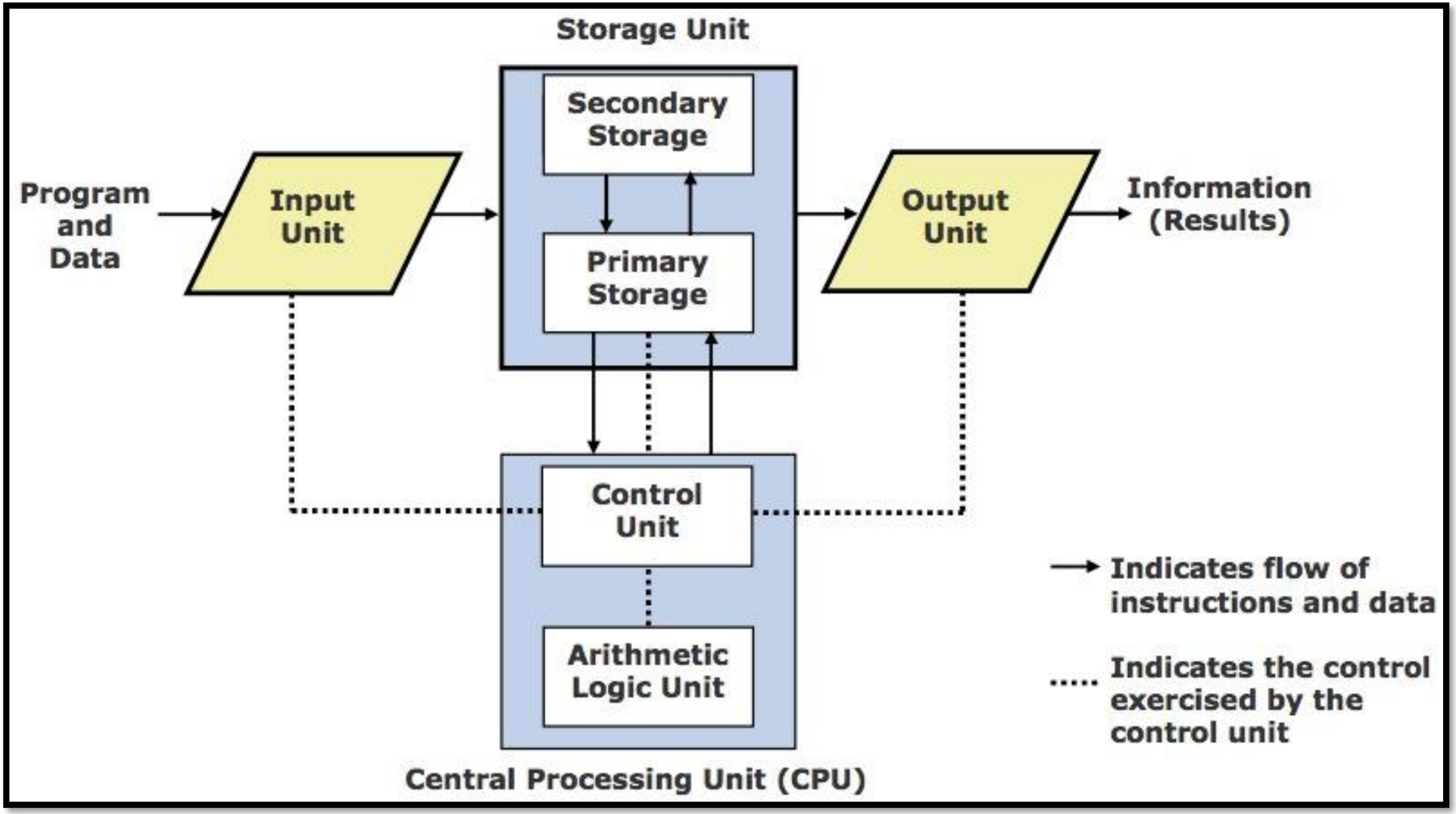
The Five Basic Operations of a Computer System

- **Inputting**. The process of **entering data** and instructions into the computer system
- **Storing**. Saving data and instructions to make them readily available for initial or additional processing whenever required
- **Processing**. Performing arithmetic operations (add, subtract, multiply, divide, etc.) or logical operations (comparisons like equal to, less than, greater than, etc.) on data to convert them into useful information
- **Outputting**. The process of producing useful information or results for the user such as a printed report or visual display
- **Controlling**. Directing the manner and sequence in which all of the above operations are performed



Basic Operations of a Computer System

Basic Organization of a Computer System



Input Unit

An input unit of a computer system performs the following functions:

① It accepts (or reads) instructions and data from outside world

② It converts these instructions and data in computer acceptable form

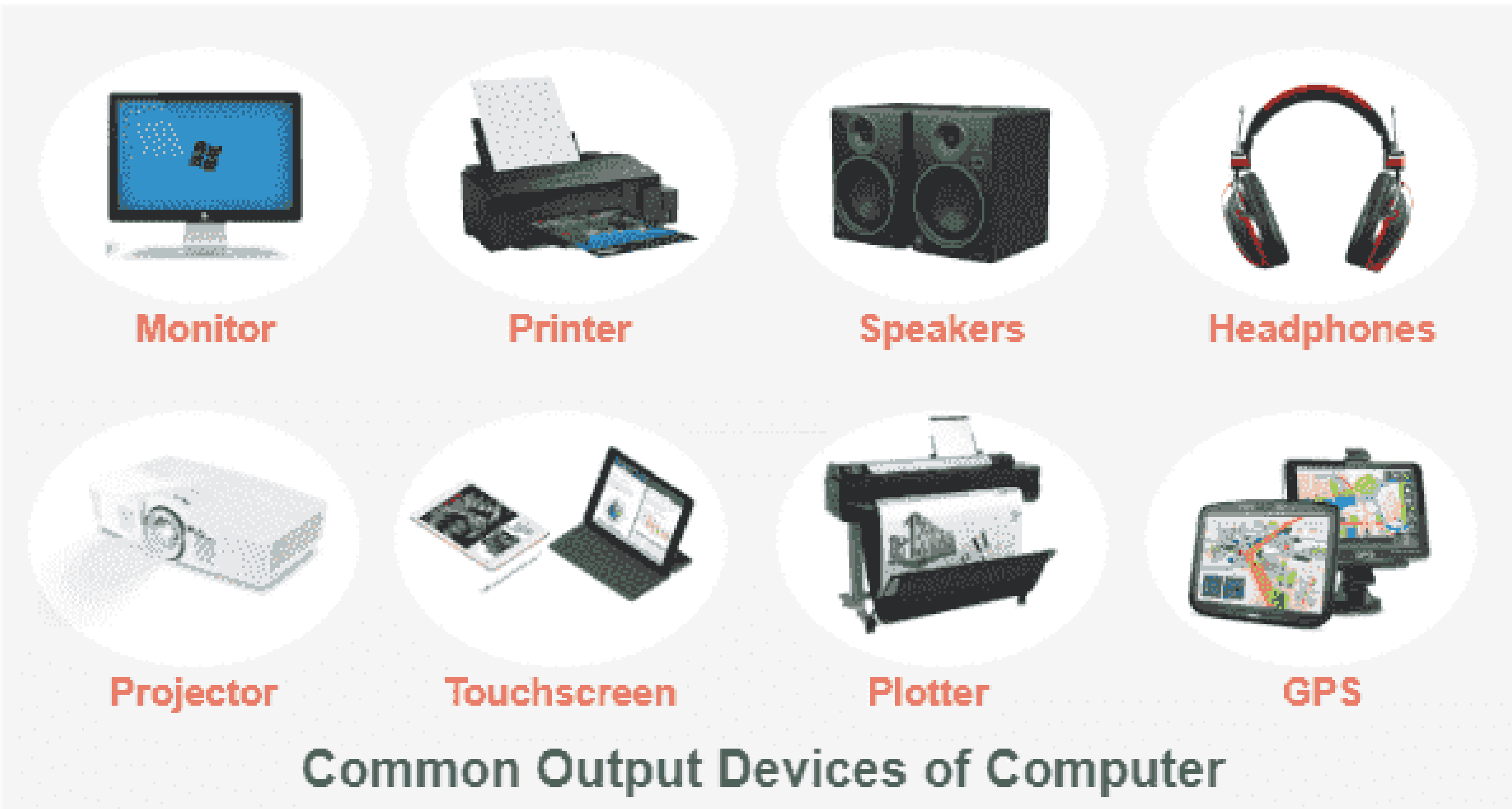
③ It supplies the converted instructions and data to the computer system for further processing

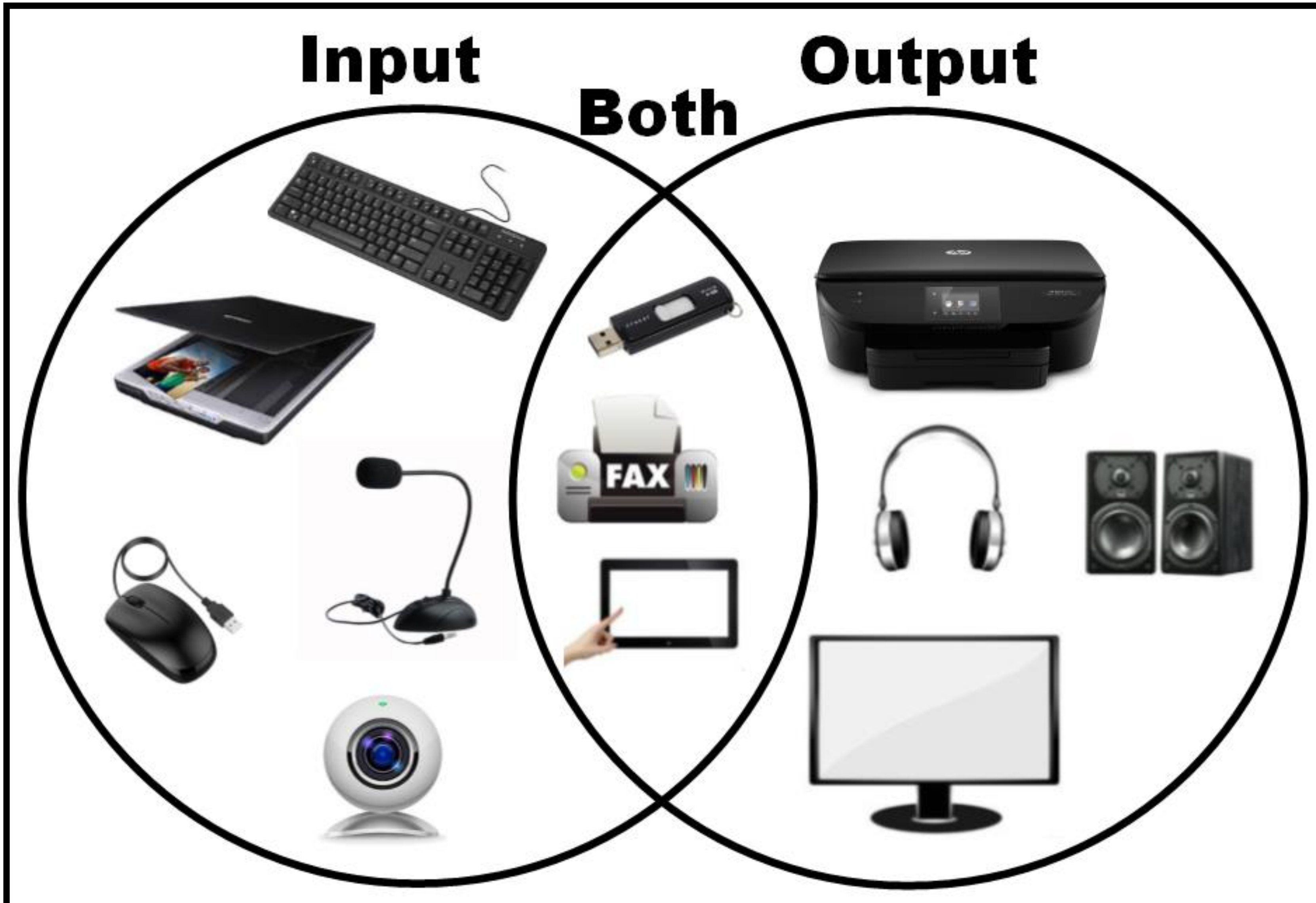

Keyboard
Mouse
Joystick
Trackball
Scanner
Touchscreen
Webcam
Microphone
Common Input Devices of Computer

Output Unit

An output unit of a computer system performs the following functions:

1. It accepts the results produced by the computer, which are in coded form and hence, cannot be easily understood by us
2. It converts these coded results to human acceptable (readable) form
3. It supplies the converted results to outside world


Monitor
Printer
Speakers
Headphones
Projector
Touchscreen
Plotter
GPS
Common Output Devices of Computer


Input
Both
Output
FAX

Storage Unit

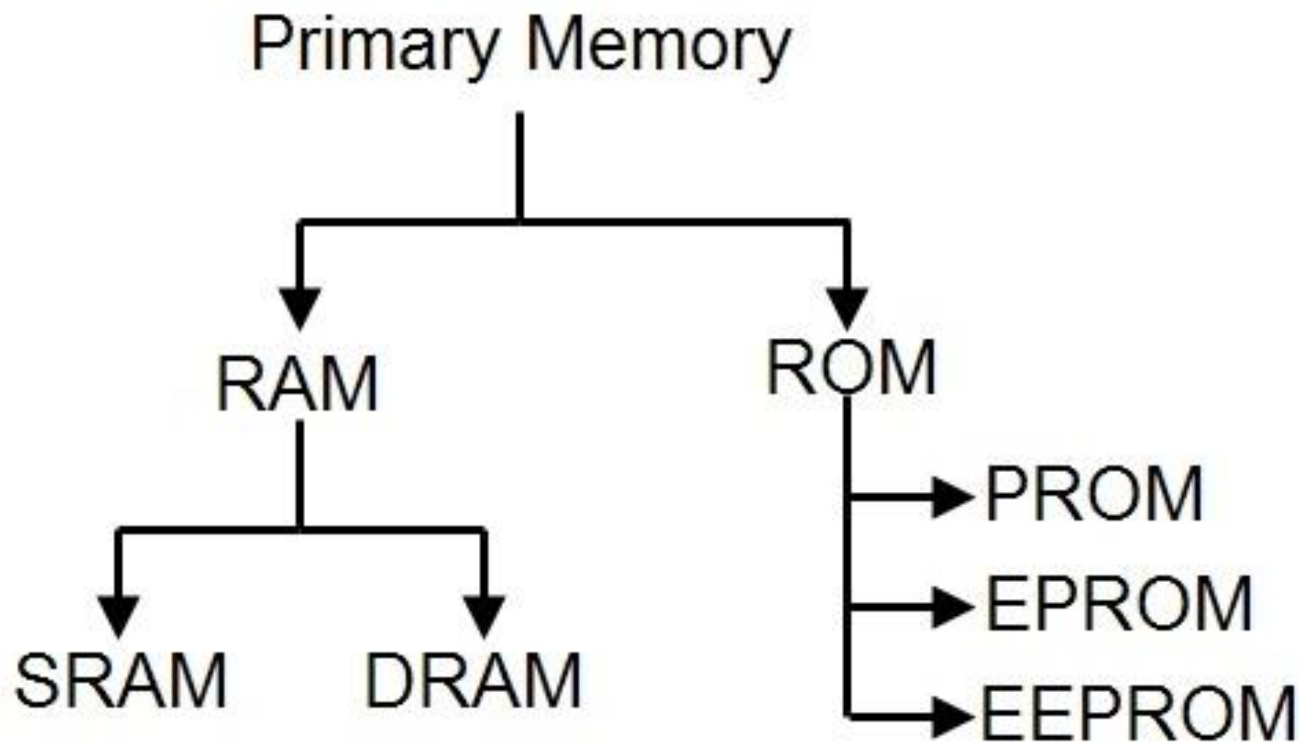
- **The storage unit of a computer system holds (or stores) the following :**
  1. Data and instructions required for processing (received from input devices)
  2. Intermediate results of processing
  3. Final results of processing, before they are released to an output device

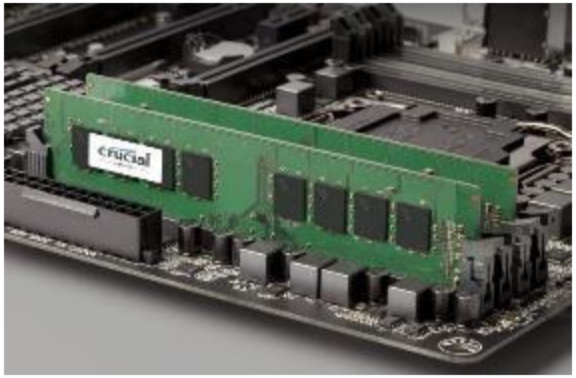

KINGSTON
FURY
BEAST
DDR4
DDR4

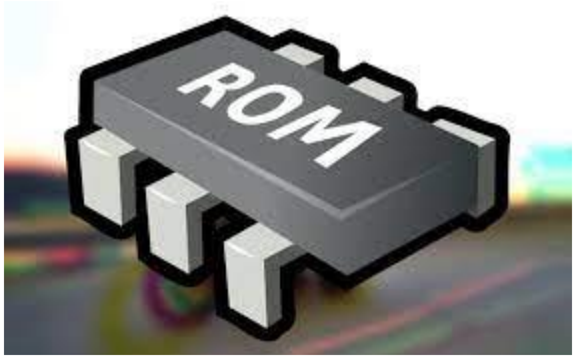
Two Types of Storage

① Primary storage

- Used to hold running program instructions
- Used to hold data, intermediate results, and results of ongoing processing of job(s)
- Fast in operation
- Small Capacity
- Expensive
- Volatile (looses data on power dissipation)


Primary Memory
RAM
ROM
SRAM
DRAM
PROM
EPROM
EEPROM




ROM

SRAM vs. DRAM

STATIC RAM	DYNAMIC RAM
Does not require refreshing	Must be continuously refreshed
Requires multiple transistors to store one bit	Requires one transistor and one capacitor to store one bit
Delivers faster access times	Delivers slower performance
Consumes less power, especially in idle	Consumes more power
Takes up more space	Requires less space
Can hold only a small amount of data	Can hold much more data
Costs more than DRAM	Costs less than SRAM
Typically used for a processor's cache	Typically used for a computer's main memory

PROM VS EPROM VS EEPROM

PROM	EPROM	EFPROM
A Read Only Memory (ROM) that can be modified only once by a users	A programmable ROM that can be erased and reused	A user-modifiable ROM that can be erased and reprogrammed repeatedly through a normal electrical voltage
Stands for Programmable Read Only Memory	Stands for Erasable Programmable Read Only Memory	Stands for Electrically Erasable Programmable Read-Only Memory
Developed by Wen Tsing Chow in 1956	Developed by Dov Frohman in 1971	Developed by George Perlegos in 1978
Reprogrammable only once	Can be reprogramed using ultraviolet light	Can be reprogramed using electrical charge

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Two Types of Storage

② Secondary storage

- Used to hold stored program instructions
- Used to hold data and information of stored jobs
- Slower than primary storage
- Large Capacity
- Lot cheaper than primary storage
- Retains data even without power

② Secondary storage

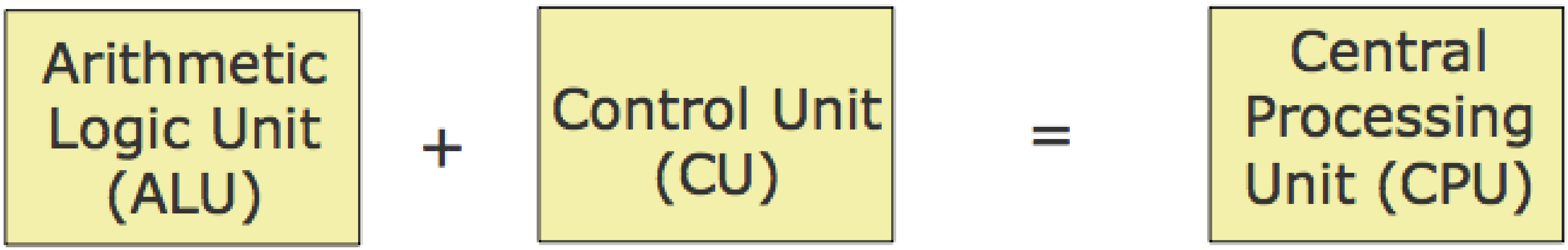
Arithmetic Logic Unit (ALU)

- Arithmetic Logic Unit of a computer system is the place where the actual executions of instructions takes place during processing operation

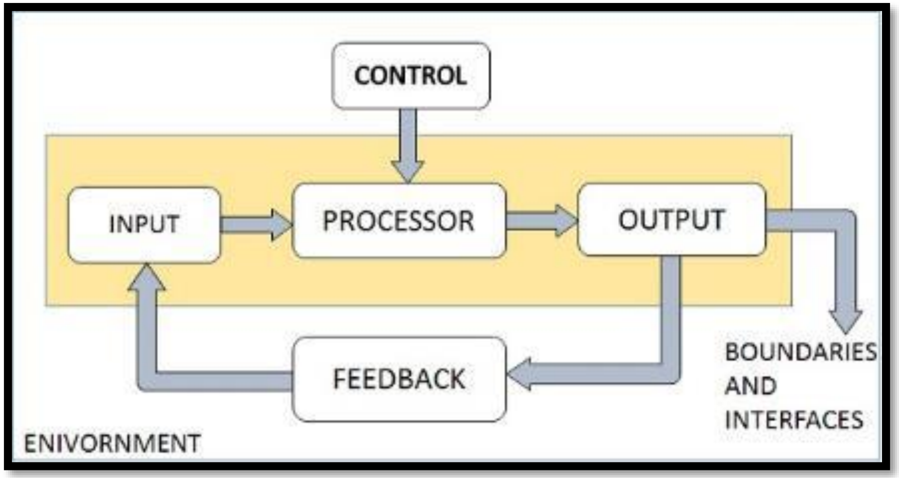
Control Unit (CU)

- Control Unit of a computer system manages and coordinates the operations of all other components of the computer system
- The control unit is a component of a computer's central processing unit (CPU) that directs operation of the processor. It controls communication and co-ordination between input/output devices. It reads and interprets instructions and determines the sequence for processing the data.
- It directs the operation of the other units by providing timing and control signals.
- All computer resources are managed by the CU (Control Unit).
- It directs the flow of data between the Central Processing Unit (CPU) and the other devices.

Central Processing Unit (CPU)



- It is the brain of a computer system
- It is responsible for controlling the operations of all other units of a computer system



The System Concept

- **A system has following three characteristics:**
  1. A system has more than one element
  2. All elements of a system are logically related
  3. All elements of a system are controlled in a manner to achieve the system goal

- A computer is a system as it comprises of integrated components (input unit, output unit, storage unit, and CPU) that work together to perform the steps called for in the executing program

Key Words/Phrases

- Arithmetic Logic Unit (ALU)
- Output interface
- Auxiliary storage
- Output unit
- Central Processing Unit (CPU)
- Outputting
- Computer system
- Primate storage
- Control Unit (CU)
- Processing
- Controlling
- Secondary storage
- Input interface
- Storage unit
- Input unit
- Storing
- Inputting
- System
- Main memory